



# 12088 - Unveiling the Nature of an Unusual Dark Matter Subhalo in the Strong Lensing System SDSSJ0946+1006 with JWST

Cycle: 5, Proposal Category: GO

## INVESTIGATORS

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## OBSERVATIONS

| <i>Folder</i>      | <i>Observation</i> | <i>Label</i> | <i>Observing Template</i> | <i>Science Target</i> |
|--------------------|--------------------|--------------|---------------------------|-----------------------|
| Observation Folder |                    |              |                           |                       |
|                    | 1                  |              | NIRCam Imaging            | (1) SDSSJ0946+1006    |

## ABSTRACT

We propose JWST/NIRCam multi-band imaging of the strong lensing system J0946+1006 to determine the nature of a previously detected low-mass subhalo. Gravitational lensing analyses of this system have revealed a subhalo that is consistent either with an anomalously compact subhalo that challenges the cold dark matter (CDM) model, or with a standard CDM subhalo hosting a faint dwarf galaxy. Current lens analyses cannot distinguish between these scenarios due to modelling degeneracies.

Our proposed observations will bypass these complexities using direct colour measurements: if the subhalo indeed hosts a dwarf galaxy, its stellar emission will show colours distinct from the lensed background source. Deep imaging with three NIRCam filter pairs will yield a decisive result: (1) Detection of a colour signal would confirm a satellite galaxy in the subhalo's center, enabling the first joint constraint on the stellar and dark matter components of an extragalactic subhalo of  $\sim 10^9$  Msun through combined lensing and stellar population synthesis. This would provide a key benchmark for galaxy formation at the lowest mass scales and establish a framework for future surveys; 2) Non-detection of subhalo light would place stringent limits on its baryonic content, implying an extreme dark halo concentration inconsistent with CDM and motivating alternatives, such as self-interacting dark matter.

This high-impact, low-risk programme leverages JWST's unprecedented imaging sensitivity and resolution in the near-infrared to deliver a definitive test of subhalo physics and the small-scale nature of dark matter.

## OBSERVING DESCRIPTION

We will use JWST/NIRCam to observe the strong lensing system SDSSJ0946+1006. The goal of this proposal is to confirm if there is a faint satellite galaxy hosted in the centre of the subhalo previously detected through gravitational lensing in this system. Below we list the pairs of filters to be used and the required exposure time:

(1) F070W + F444W: 5862 s

(2) F115W + F356W: 3307 s

(3) F200W + F277W: 2029 s

These observations will ensure that, even in the worst cases, a galaxy hosted in the subhalo's center (if present) will be detected with  $\text{SNR} > 5$  in all bands. The combinations of the SW+LW filters are chosen to avoid possible saturation from the foreground lens galaxy. To ensure high-quality imaging, we require seven dithers for each exposure.

# Proposal 12088 - Targets - Unveiling the Nature of an Unusual Dark Matter Subhalo in the Strong Lensing System SDSSJ0946+1006 ...

| Fixed Targets | #                                                                                                                 | Name           | Target Coordinates               | Targ. Coord. Corrections | Miscellaneous |
|---------------|-------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------|--------------------------|---------------|
|               | (1)                                                                                                               | SDSSJ0946+1006 | RA: 09 46 56.6800 (146.7361667d) |                          |               |
|               |                                                                                                                   |                | Dec: +10 06 52.80 (10.11467d)    |                          |               |
|               |                                                                                                                   |                | Equinox: J2000                   |                          |               |
|               | Comments:<br>Category=Galaxy<br>Description=[Dwarf galaxies, Einstein rings, Elliptical galaxies]<br>Extended=YES |                |                                  |                          |               |

# Proposal 12088 - Observation 1 - Unveiling the Nature of an Unusual Dark Matter Subhalo in the Strong Lensing System SDSSJ0946...

|                      |                                                                                                                                               |                     |                                                                                     |                 |                          |                      |                    |               |                     |                    |  |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|-----------------|--------------------------|----------------------|--------------------|---------------|---------------------|--------------------|--|
| Observation          | <b>Proposal 12088, Observation 1</b> <span style="float: right;">Mon Jun 08 16:00:17 GMT 2026</span>                                          |                     |                                                                                     |                 |                          |                      |                    |               |                     |                    |  |
|                      | <b>Diagnostic Status: Warning</b><br>Observing Template: NIRCam Imaging                                                                       |                     |                                                                                     |                 |                          |                      |                    |               |                     |                    |  |
| Diagnostics          | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                   |                     |                                                                                     |                 |                          |                      |                    |               |                     |                    |  |
| Fixed Targets        | #                                                                                                                                             | Name                | Target Coordinates                                                                  |                 | Targ. Coord. Corrections |                      |                    | Miscellaneous |                     |                    |  |
|                      | (1)                                                                                                                                           | SDSSJ0946+1006      | RA: 09 46 56.6800 (146.7361667d)<br>Dec: +10 06 52.80 (10.11467d)<br>Equinox: J2000 |                 |                          |                      |                    |               |                     |                    |  |
| Template             | <i>Comments:</i><br><i>Category=Galaxy</i><br><i>Description=[Dwarf galaxies, Einstein rings, Elliptical galaxies]</i><br><i>Extended=YES</i> |                     |                                                                                     |                 |                          |                      |                    |               |                     |                    |  |
|                      | Module                                                                                                                                        |                     |                                                                                     |                 |                          | Subarray             |                    |               |                     |                    |  |
| Dithers              | B                                                                                                                                             |                     |                                                                                     |                 |                          | FULL                 |                    |               |                     |                    |  |
|                      | #                                                                                                                                             | Primary Dither Type |                                                                                     | Primary Dithers |                          | Subpixel Dither Type |                    | Dither Size   |                     | Subpixel Positions |  |
| Spectral Elements    | 1                                                                                                                                             | INTRASCA            |                                                                                     | 7               |                          | STANDARD             |                    | 8" (SMALL)    |                     | 1                  |  |
|                      | #                                                                                                                                             | Short Filter        | Long Filter                                                                         | Readout Pattern | Groups/Int               | Integrations/Exp     | Total Integrations | Total Dithers | Total Exposure Time | Optional ETC ID    |  |
|                      | 1                                                                                                                                             | F070W               | F444W                                                                               | MEDIUM8         | 8                        | 1                    | 7                  | 7             | 5862.276            | 276667             |  |
|                      | 2                                                                                                                                             | F115W               | F356W                                                                               | SHALLOW4        | 9                        | 1                    | 7                  | 7             | 3306.925            | 276667             |  |
| Special Requirements | 3                                                                                                                                             | F200W               | F277W                                                                               | SHALLOW2        | 6                        | 1                    | 7                  | 7             | 2029.25             | 276667             |  |
|                      | Offset 40.0 arcsec, 40.0 arcsec                                                                                                               |                     |                                                                                     |                 |                          |                      |                    |               |                     |                    |  |